

A Crash Course on Bayesian Statistics and its Philosophy

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September 1, 2026

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1 The Bayesian Recipe

Statistical inference is concerned with learning about unknown quantities from observed data.

In many problems, we model the observed data as being generated from a probability distribution P_θ where $\theta \in \Theta$. Thus, we have a class of probability distributions indexed by a parameter space Θ . The problem is to infer about the underlying θ after observing data. This abstraction is the starting point for simple parametric models all the way up to state of the art ML models like Large Language Models and Diffusion Models.

The Bayesian approach begins with a simple idea.

Before seeing any data, we are uncertain about the value of θ . This uncertainty itself should be represented using probability.

Thus we place a probability distribution on the parameter space Θ . This distribution is called the *prior distribution* and is denoted by

$$\pi(\theta).$$

The data are modeled through a conditional distribution

$$p(x \mid \theta),$$

called the *likelihood*.

After observing data

$$X = x,$$

Bayes' rule gives the posterior distribution

$$\pi(\theta \mid x) = \frac{p(x \mid \theta)\pi(\theta)}{\int_{\Theta} p(x \mid u)\pi(u) du}.$$

The posterior distribution summarizes what we believe about θ after observing the data. The numerator is simply the product of the prior and the likelihood of the observed data, while the denominator is the marginal probability of the observed data, often called evidence in the literature.

$\text{Posterior} = \frac{\text{Likelihood} \times \text{Prior}}{\text{Evidence}}.$

Bayesian inference can therefore be summarized in three steps:

1. Specify a prior distribution (Modelling Step)
2. Specify a likelihood (Modelling Step).
3. Apply Bayes' rule.

Everything else in Bayesian statistics follows from these three steps.

2 Examples of Bayesian Inference

Example 2.1 (A Patient Covered in Spots). Suppose a patient arrives at a hospital covered in spots. The symptoms are consistent with either chickenpox or smallpox.

Let

$$\Theta \in \{\text{smallpox}, \text{chickenpox}, \text{neither}\}.$$

Suppose we assign the likelihoods

$$P(\text{spots} \mid \text{smallpox}) = 0.9,$$

$$P(\text{spots} \mid \text{chickenpox}) = 0.8,$$

and

$$P(\text{spots} \mid \text{neither}) = 0.$$

Further suppose that smallpox is extremely rare and we assign the prior

$$P(\text{smallpox}) = 0.001,$$

while

$$P(\text{chickenpox}) = 0.1.$$

This leaves

$$P(\text{neither}) = 0.899.$$

The inference question is:

$$P(\text{smallpox} \mid \text{spots})$$

and

$$P(\text{chickenpox} \mid \text{spots}).$$

To calculate these probabilities, we first compute the overall probability of observing spots.

Using the law of total probability,

$$P(\text{spots}) = P(\text{spots} \mid \text{smallpox})P(\text{smallpox}) + P(\text{spots} \mid \text{chickenpox})P(\text{chickenpox}) + P(\text{spots} \mid \text{neither})P(\text{neither}).$$

Substituting the numerical values gives

$$P(\text{spots}) = 0.9(0.001) + 0.8(0.1) + 0(0.899) = 0.0809.$$

We now apply Bayes' rule.

For smallpox,

$$P(\text{smallpox} \mid \text{spots}) = \frac{P(\text{spots} \mid \text{smallpox})P(\text{smallpox})}{P(\text{spots})}.$$

Substituting the values,

$$P(\text{smallpox} \mid \text{spots}) = \frac{0.9 \times 0.001}{0.0809} \approx 0.011.$$

Similarly,

$$P(\text{chickenpox} \mid \text{spots}) = \frac{P(\text{spots} \mid \text{chickenpox})P(\text{chickenpox})}{P(\text{spots})} = \frac{0.8 \times 0.1}{0.0809} \approx 0.989.$$

Thus

$$P(\text{smallpox} \mid \text{spots}) \approx 0.011,$$

while

$$P(\text{chickenpox} \mid \text{spots}) \approx 0.989.$$

Even though spots are slightly more likely under smallpox than under chickenpox, smallpox is so rare that the posterior probability of chickenpox is overwhelmingly larger.

The Bayesian conclusion is therefore:

The patient almost certainly has chickenpox rather than smallpox.

Remark 2.2. *Notice what happened here.*

The likelihood slightly favors smallpox because

$$P(\text{spots} \mid \text{smallpox}) = 0.9 > 0.8 = P(\text{spots} \mid \text{chickenpox}).$$

However, the prior strongly favors chickenpox because

$$P(\text{chickenpox}) = 0.1 \gg 0.001 = P(\text{smallpox}).$$

Bayes' rule combines these two sources of information.

This example illustrates a central lesson of Bayesian statistics:

The most likely explanation of the data need not be the most plausible explanation after taking prior information into account.

2.1 Contrast with Maximum Likelihood Estimation

A widely used alternative to full Bayesian inference is *maximum likelihood estimation*.

Given observed data x , the maximum likelihood estimator is defined by

$$\hat{\theta}_{\text{MLE}} = \arg \max_{\theta} p(x \mid \theta).$$

In words, the maximum likelihood estimator chooses the parameter value that makes the observed data most likely.

Returning to the spots example, the observed data are simply that the patient has spots. The likelihoods are

$$P(\text{spots} \mid \text{smallpox}) = 0.9,$$

and

$$P(\text{spots} \mid \text{chickenpox}) = 0.8.$$

Since

$$0.9 > 0.8,$$

the maximum likelihood estimator chooses

$$\hat{\theta}_{\text{MLE}} = \text{smallpox}.$$

This conclusion is very different from the Bayesian conclusion.

The Bayesian analysis gave

$$P(\text{smallpox} \mid \text{spots}) \approx 0.011,$$

and

$$P(\text{chickenpox} \mid \text{spots}) \approx 0.989.$$

Thus the Bayesian analysis overwhelmingly favors chickenpox.

The reason for this discrepancy is simple.

Maximum likelihood only uses the likelihood:

$$p(x \mid \theta).$$

Bayesian inference uses both the likelihood and the prior:

$$\pi(\theta \mid x) \propto p(x \mid \theta)\pi(\theta).$$

The prior incorporates the fact that smallpox is extraordinarily rare, while maximum likelihood ignores this information entirely.

Conceptually, the two approaches answer different questions.

Maximum likelihood asks:

Which parameter value makes the observed data most likely?

Bayesian inference asks:

Given the observed data, what should we believe about the parameter?

In many problems the answers may be similar. In this example, however, the answers are dramatically different.

This example highlights one of the major strengths of Bayesian inference: it provides a principled way of incorporating prior information into statistical decision making.

Example 2.3 (Six Measurements). Suppose a scientist measures an unknown physical quantity θ six times and obtains

$$26.6, 38.5, 34.4, 34, 31, 23.6.$$

Denote these observations by

$$x_1, \dots, x_n,$$

where

$$n = 6.$$

On the basis of these measurements, what can we infer about θ ?

The Bayesian solution begins with a probabilistic model. We need to specify both a likelihood and a prior distribution.

The likelihood represents the probability of the observed data given parameter values. Besides the unknown quantity θ , we introduce another parameter $\sigma > 0$ which represents the scale of the measurement noise.

Thus our parameter vector is

$$(\theta, \sigma).$$

We shall use the Gaussian likelihood

$$L(\theta, \sigma) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi}\sigma} \exp \left\{ -\frac{(x_i - \theta)^2}{2\sigma^2} \right\}.$$

At this point it is worth discussing carefully what this likelihood means.

Let

$$X_1, \dots, X_n$$

denote potential measurements. Because measurements are recorded only up to some finite precision, the observed value 26.6 should really be interpreted as belonging to a small interval

$$[26.6 - \delta, 26.6 + \delta]$$

for some small rounding error $\delta > 0$.

Thus the likelihood is formally

$$L(\theta, \sigma) = P\left(X_1 \in [x_1 - \delta, x_1 + \delta], \dots, X_n \in [x_n - \delta, x_n + \delta] \mid \theta, \sigma\right).$$

For small δ , a probability-density approximation gives

$$L(\theta, \sigma) \approx \delta^n f_{X_1, \dots, X_n \mid \theta, \sigma}(x_1, \dots, x_n).$$

We now assume that

$$f_{X_1, \dots, X_n \mid \theta, \sigma}(x_1, \dots, x_n) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi}\sigma} \exp \left\{ -\frac{(x_i - \theta)^2}{2\sigma^2} \right\}.$$

The multiplicative factor δ^n is constant with respect to θ and σ , so it can be ignored. This leads to the likelihood above.

A subtle point is worth emphasizing here.

Many texts write the model as

$$X_1, \dots, X_n \mid \theta, \sigma \stackrel{\text{i.i.d.}}{\sim} N(\theta, \sigma^2).$$

This is a stronger statement than simply specifying the likelihood at the observed data.

Indeed, the i.i.d. normal assumption means that for *every*

$$(u_1, \dots, u_n) \in \mathbb{R}^n,$$

the joint density satisfies

$$f_{X_1, \dots, X_n | \theta, \sigma}(u_1, \dots, u_n) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi}\sigma} \exp \left\{ -\frac{(u_i - \theta)^2}{2\sigma^2} \right\}.$$

The posterior calculation only uses the likelihood evaluated at the observed data. The full i.i.d. normal assumption specifies the probability law of all possible datasets that could have been observed. The following remark expands on this point.

Remark 2.4 (Likelihood versus full probabilistic model). *It is important to distinguish between the likelihood and a full probabilistic model for the data.*

The assumption

$$X_1, \dots, X_n \mid \theta, \sigma \stackrel{\text{i.i.d.}}{\sim} N(\theta, \sigma^2)$$

is equivalent to requiring that

$$f_{X_1, \dots, X_n | \theta, \sigma}(u_1, \dots, u_n) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi}\sigma} \exp \left\{ -\frac{(u_i - \theta)^2}{2\sigma^2} \right\}$$

for every possible dataset

$$(u_1, \dots, u_n) \in \mathbb{R}^n.$$

This is a very strong assumption.

Indeed, consider the alternative model

$$f_{X_1, \dots, X_n | \theta, \sigma}(u_1, \dots, u_n) = \begin{cases} \prod_{i=1}^n \frac{1}{\sqrt{2\pi}\sigma} \exp \left\{ -\frac{(u_i - \theta)^2}{2\sigma^2} \right\}, & (u_1, \dots, u_n) \in (20, 30)^n, \\ \text{arbitrary}, & (u_1, \dots, u_n) \notin (20, 30)^n. \end{cases}$$

This model is clearly not the same as the i.i.d. normal model.

However, if the observed data happen to be

$$(x_1, \dots, x_n) \in (20, 30)^n,$$

then the likelihood function obtained from this alternative model is exactly the same as the Gaussian likelihood.

Consequently, the Bayesian posterior calculation would be identical.

This illustrates that the posterior depends only on the likelihood evaluated at the observed data and not on the full probability model for all possible datasets.

To complete the Bayesian model, we need a prior distribution.

We take

$$\theta \sim \text{Uniform}(-C, C)$$

and

$$\log \sigma \sim \text{Uniform}(-C, C),$$

independently, where C is a very large positive constant.

The idea is that we do not wish to express strong prior information about either parameter.

The corresponding joint prior density is

$$\pi(\theta, \sigma) \propto \frac{\mathbf{1}\{-C < \theta < C, -C < \log \sigma < C\}}{\sigma}.$$

Using Bayes' rule,

$$\pi(\theta, \sigma \mid x_1, \dots, x_n) \propto \pi(\theta, \sigma) L(\theta, \sigma).$$

Substituting the prior and likelihood gives

$$\pi(\theta, \sigma \mid x_1, \dots, x_n) \propto \sigma^{-n-1} \exp \left\{ -\frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \theta)^2 \right\} \mathbf{1}\{-C < \theta < C, -C < \log \sigma < C\}.$$

This is the joint posterior density of θ and σ .

To obtain the posterior density of θ , we integrate out σ :

$$\pi(\theta \mid x_1, \dots, x_n) \propto \mathbf{1}\{-C < \theta < C\} \int_{e^{-C}}^{e^C} \sigma^{-n-1} \exp \left\{ -\frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \theta)^2 \right\} d\sigma.$$

Since C is very large, we may replace the limits of integration by 0 and ∞ .

Define

$$S(\theta) = \sum_{i=1}^n (x_i - \theta)^2.$$

Then

$$\pi(\theta \mid x_1, \dots, x_n) \propto \int_0^\infty \sigma^{-n-1} \exp \left\{ -\frac{S(\theta)}{2\sigma^2} \right\} d\sigma.$$

Using the change of variables

$$t = \frac{S(\theta)}{2\sigma^2},$$

one obtains

$$\pi(\theta \mid x_1, \dots, x_n) \propto [S(\theta)]^{-n/2}.$$

Thus the posterior density is inversely proportional to a power of the residual sum of squares. Let

$$\hat{\theta} = \bar{x} = \frac{1}{n} \sum_{i=1}^n x_i.$$

Since $\hat{\theta}$ minimizes $S(\theta)$, the posterior density is maximized at $\hat{\theta}$. Using the standard least-squares identity,

$$S(\theta) = S(\hat{\theta}) + n(\theta - \hat{\theta})^2,$$

we obtain

$$\pi(\theta \mid x_1, \dots, x_n) \propto \left(\frac{1}{S(\hat{\theta}) + n(\theta - \hat{\theta})^2} \right)^{n/2}.$$

Equivalently,

$$\pi(\theta \mid x_1, \dots, x_n) \propto \left(1 + \frac{n(\theta - \hat{\theta})^2}{S(\hat{\theta})} \right)^{-n/2}.$$

This is exactly a Student- t density. In fact,

$$\frac{\sqrt{n}(\theta - \hat{\theta})}{\sqrt{S(\hat{\theta})/(n-1)}} \mid x_1, \dots, x_n \sim t_{n-1}.$$

Thus the posterior distribution is centered at the sample mean. The posterior mean, median and mode are all equal to

$$\hat{\theta} = \bar{x}.$$

A $100(1 - \alpha)\%$ Bayesian credible interval for θ is therefore

$$\left[\hat{\theta} - \frac{t_{n-1, \alpha/2}}{\sqrt{n}} \sqrt{\frac{S(\hat{\theta})}{n-1}}, \hat{\theta} + \frac{t_{n-1, \alpha/2}}{\sqrt{n}} \sqrt{\frac{S(\hat{\theta})}{n-1}} \right].$$

For $\alpha = 0.05$, this gives the interval

$$[25.598, 37.102].$$

Remark 2.5 (Bayesian versus Frequentist Interpretation). *An interesting feature of this example is that the Bayesian credible interval is numerically identical to the classical Student- t confidence*

interval. However, the interpretations are different.

The frequentist interpretation is:

If we repeated the experiment many times and constructed a confidence interval each time, then approximately 95% of those intervals would contain the true value of θ .

The Bayesian interpretation is:

Given the observed data, the posterior probability that θ lies in the interval is 0.95.

2.2 Frequentist and Bayesian Inference Can Disagree

In the previous example, the Bayesian credible interval and the frequentist confidence interval happened to coincide. The agreement in the previous example is nice but coincidental and it is easy to break this coincidence. A slight change in the experimental design can lead to different conclusions.

Consider the following modification of the measurement problem.

Example 2.6 (Measurements with a Stopping Rule). A scientist repeatedly measures an unknown physical quantity θ until a measurement smaller than 25 is obtained.

Suppose the resulting measurements are

$$26.6, ; 38.5, ; 34.4, ; 34, ; 31, ; 23.6.$$

Should our inference about θ change?

Notice that the observed data are exactly the same as in the previous example. The only difference is the manner in which the data were collected.

2.2.1 Frequentist Analysis

The confidence interval from the previous example is no longer automatically valid. The reason is that the sample size is no longer fixed. Instead, the number of observations becomes the random variable

$$N = \inf\{n \geq 1 : X_n \leq 25\}. \tag{1}$$

The confidence interval from the previous example was based on the probability statement

$$\frac{\sqrt{n}(\bar{X}_n - \theta)}{\hat{\sigma}} \sim t_{n-1},$$

which assumes that the sample size is deterministically equal to n .

In the present problem, the relevant probability becomes

$$P \left(\bar{X}_N - \frac{t_{N-1, \alpha/2}}{\sqrt{N}} \sqrt{\frac{1}{N-1} \sum_{i=1}^N (X_i - \bar{X}_N)^2} \leq \theta \leq \bar{X}_N + \frac{t_{N-1, \alpha/2}}{\sqrt{N}} \sqrt{\frac{1}{N-1} \sum_{i=1}^N (X_i - \bar{X}_N)^2} \right).$$

Here the probability is computed under the model

$$X_1, X_2, \dots, X_N \stackrel{\text{i.i.d.}}{\sim} N(\theta, \sigma^2),$$

where N is the random variable in (1).

There is no reason for the probability above to be exactly equal to $1 - \alpha$. In other words, the stopping rule changes the repeated sampling distribution of the procedure.

Constructing valid frequentist confidence intervals in the presence of stopping rules is a subtle problem and remains an active area of research.

2.2.2 Bayesian Analysis

In contrast to frequentist inference, the Bayesian inference procedure will not change. This is because the likelihood function in the present problem is exactly the same as the likelihood function in the original fixed-sample-size problem.

To verify this, let δ denote the (very small) rounding error in the measurements. The likelihood in the present problem is

$$\begin{aligned} \text{Likelihood} &= P(\text{observed data} \mid \theta, \sigma) \\ &= P(X_1 \in [x_1 - \delta, x_1 + \delta], \dots, X_6 \in [x_6 - \delta, x_6 + \delta], \quad X_1 \geq 25, \dots, X_5 \geq 25, X_6 < 25 \mid \theta, \sigma). \end{aligned}$$

In other words, we are writing the probability that the first five observations are all larger than 25 while the sixth observation is smaller than 25, in addition to the exact observed values.

However, these additional constraints do not change the probability.

As a simple example, note that $P(Z = 5) = P(Z = 5, Z < 10)$. The additional restriction $Z < 10$ does not affect the probability because it is already implied by the event $Z = 5$.

Similarly,

$$\begin{aligned} \text{Likelihood} &= P(X_1 \in [x_1 - \delta, x_1 + \delta], \dots, X_6 \in [x_6 - \delta, x_6 + \delta], \\ &\quad X_1 \geq 25, \dots, X_5 \geq 25, X_6 < 25 \mid \theta, \sigma) = P(X_1 \in [x_1 - \delta, x_1 + \delta], \dots, X_6 \in [x_6 - \delta, x_6 + \delta] \mid \theta, \sigma). \end{aligned}$$

But the expression on the right-hand side is exactly the likelihood from the original problem.

Therefore,

Likelihood in Problem 1 = Likelihood in Problem 2.

Since the likelihood is the same in both problems, and since there is no reason to change the prior distribution, the posterior distribution is also exactly the same.

Consequently, all Bayesian conclusions remain unchanged. The posterior mean, posterior mode, posterior credible interval, and indeed the entire posterior distribution are identical to those obtained in the original problem.

Therefore, from the Bayesian perspective, stopping rules can be ignored for inference about θ , because they do not affect the likelihood.

2.2.3 The Likelihood Principle

The likelihood principle states that all evidence in the data relevant to the unknown parameters is contained in the likelihood function. The two problems considered above have exactly the same likelihood function. A Bayesian therefore reaches exactly the same conclusion in both cases. A frequentist, on the other hand, generally reaches different conclusions because frequentist procedures depend not only on the observed data but also on the sampling plan and stopping rule. Thus standard frequentist inference violates the likelihood principle, whereas Bayesian inference automatically satisfies it.

3 Coin Fairness Testing and the Likelihood Principle

We now consider another example where frequentist and Bayesian reasoning can lead to different conclusions. The example concerns testing whether a coin is fair.

3.1 The Problem

Suppose a coin is tossed 12 times and the observed sequence is

TTTTHTHTTTTH.

Thus there are 3 heads and 9 tails.

The question is:

Is there evidence that the coin is biased against heads?

Let p denote the probability that the coin lands heads. The null hypothesis of fairness is

$$H_0 : p = \frac{1}{2}.$$

A natural one-sided alternative is

$$H_1 : p < \frac{1}{2}.$$

We now compare two frequentist analyses of exactly the same observed data.

3.2 Frequentist Solution: Fixed Sample Size

First suppose that the experiment was designed in advance as follows:

Toss the coin exactly 12 times.

Let X denote the number of heads in 12 tosses. Under the null hypothesis $H_0 : p = 1/2$, we have

$$X \sim \text{Binomial}\left(12, \frac{1}{2}\right).$$

The observed value is

$$X_{\text{obs}} = 3.$$

To test against the alternative $p < 1/2$, the p-value is the probability, under the null hypothesis, of observing 3 or fewer heads:

$$\text{p-value} = P_{p=1/2}(X \leq 3).$$

Therefore

$$\begin{aligned} P_{p=1/2}(X \leq 3) &= \sum_{k=0}^3 \binom{12}{k} \left(\frac{1}{2}\right)^{12} \\ &= \left[\binom{12}{0} + \binom{12}{1} + \binom{12}{2} + \binom{12}{3} \right] \left(\frac{1}{2}\right)^{12} \\ &= \frac{1 + 12 + 66 + 220}{4096} \\ &= \frac{299}{4096} \\ &\approx 0.073. \end{aligned}$$

Thus the p-value is approximately 7.3%. At the usual 5% significance level, we would not reject the null hypothesis that the coin is fair.

3.3 Frequentist Solution: Stopping After Three Heads

Now suppose instead that the experiment was designed differently:

Toss the coin until three heads have appeared.

The observed sequence is still

$$TTTTHTHTTTTH,$$

so the third head appeared on the 12th toss.

Let N denote the number of tosses needed to obtain three heads. Under the null hypothesis $p = 1/2$, N has a negative binomial distribution.

The observed value is

$$N_{\text{obs}} = 12.$$

Evidence against fairness in the direction $p < 1/2$ corresponds to needing an unusually large number of tosses to obtain three heads. Therefore the p-value is

$$P_{p=1/2}(N \geq 12).$$

Equivalently,

$$P_{p=1/2}(N \geq 12) = 1 - P_{p=1/2}(N \leq 11).$$

For $n \geq 3$, the probability that the third head occurs exactly on toss n is

$$P_{p=1/2}(N = n) = \binom{n-1}{2} \left(\frac{1}{2}\right)^n.$$

Indeed, among the first $n - 1$ tosses there must be exactly two heads, and the n th toss must be a head.

Thus

$$\begin{aligned} P_{p=1/2}(N \geq 12) &= 1 - \sum_{n=3}^{11} \binom{n-1}{2} \left(\frac{1}{2}\right)^n \\ &= \frac{134}{4096} \\ &\approx 0.0327. \end{aligned}$$

So in this second analysis the p-value is approximately 3.27%. At the usual 5% significance level, we would reject the null hypothesis that the coin is fair.

3.4 What Changed?

The observed sequence of tosses is exactly the same in the two analyses. In both cases we observed

$$TTTTHTHTTTTH,$$

which contains 3 heads and 9 tails.

However, the frequentist p-value changed from

$$0.073$$

to

$$0.0327.$$

The reason is that frequentist p-values are computed using the sampling distribution of the test statistic under the null hypothesis. This sampling distribution depends not only on the observed data, but also on the experimental design.

If the number of tosses was fixed in advance, the relevant null distribution is binomial. If the experiment was stopped after three heads appeared, the relevant null distribution is negative binomial.

Thus, under frequentist testing, the conclusion depends on the stopping rule.

Remark 3.1. *The sample space changed, in the first case, the sample space is the set of sequences of H and T of length 12. In the second case, the sample space is the set of sequences of arbitrary length finishing with the third H. Probability assignment in the sample space has also naturally changed. When we calculate p values, we are calculating probabilities of the observed sequence plus some other hypothetical sequences that we have not observed. This is the reason the p value changed.*

3.5 Bayesian Solution

From the Bayesian perspective, the inference does not depend on whether the experimenter decided in advance to toss the coin 12 times or instead decided to continue tossing until three heads were observed.

The reason is that the likelihood function is identical in both situations.

In the first experiment, the observed data consist of three heads and nine tails. Ignoring multiplicative constants that do not depend on p , the likelihood is

$$L(p) \propto p^3(1-p)^9.$$

In the second experiment, the coin is tossed until the third head appears. The probability of observing the particular sequence

$$TTTTHTHTTTTH$$

is again proportional to

$$L(p) \propto p^3(1-p)^9.$$

Thus the two experiments lead to exactly the same likelihood function.

If the prior distribution for p is also taken to be the same in both experiments, then Bayes' rule gives

$$\pi(p \mid \text{data}) \propto L(p)\pi(p),$$

and therefore the posterior distribution is identical in the two cases.

Consequently, all Bayesian conclusions are unchanged: the posterior mean, posterior mode, credible intervals, posterior probabilities, and any Bayesian decision rule will be exactly the same.

This is another illustration of the likelihood principle. The two experiments produce the same likelihood and therefore a Bayesian reaches the same conclusion in both cases. The frequentist p-value, however, changes from 7.3% to 3.27% because the sampling plan changes. Thus standard p-value based inference depends on aspects of the experiment beyond the observed likelihood, whereas Bayesian inference depends only on the likelihood and the prior.

3.6 The Central Distinction: Observed versus Hypothetical Data

The preceding example illustrates a fundamental difference between Bayesian and frequentist inference.

A useful way to remember the distinction is the following:

Bayesian inference asks: How probable is the data I actually observed, as a function of the unknown parameter? Frequentist inference asks: How would my procedure behave over data sets that I could have observed?

This distinction explains why stopping rules can affect frequentist inference even when they do not affect Bayesian inference.

In the coin-tossing example, the data that were actually observed are the same under both experimental designs: we observed a particular sequence containing three heads and nine tails. Consequently, under either design, the likelihood for p is proportional to

$$p^3(1-p)^9.$$

Thus, if the prior distribution is the same, the posterior distribution is also the same.

The frequentist calculation is different. A p-value is not determined only by the probability of the data that were actually observed. It also involves probabilities of other, hypothetical data sets that could have been observed under the experimental design.

Under the fixed-sample-size design, these hypothetical data sets are sequences of exactly 12 tosses. Under the stopping-rule design, they are sequences of variable length that terminate when the third head is observed. The collection of possible data sets has therefore changed, and so has

the relevant sampling distribution. This is why the frequentist p -value can change even though the observed data have not.

The phenomenon can be summarized as

A stopping rule changes the data sets that *could have been observed*, but it need not change the likelihood of the data that *were actually observed*.

This is the essential reason that stopping rules can matter for frequentist inference while being irrelevant for Bayesian inference when the stopping rule contributes no parameter-dependent information to the likelihood.

3.7 A striking example: will the coin ever produce a head?

Consider a coin whose probability of heads is p . Suppose first that we place the uniform prior

$$p \sim \text{Unif}(0, 1),$$

and observe n consecutive tails.

By Beta–Binomial conjugacy,

$$p \mid T^n \sim \text{Beta}(1, n + 1).$$

Thus the posterior is concentrating near zero. In fact, the posterior mean is

$$\mathbb{E}[p \mid T^n] = \frac{1}{n + 2},$$

so after a long sequence of tails we become increasingly convinced that p is very small.

Now ask a seemingly natural predictive question:

What is the posterior predictive probability that the next n tosses also contain no heads?

Conditional on p , this probability is $(1 - p)^n$. Hence

$$\begin{aligned} \mathbb{P}(T_{\text{next}}^n \mid T^n) &= \int_0^1 (1 - p)^n (n + 1) (1 - p)^n dp \\ &= (n + 1) \int_0^1 (1 - p)^{2n} dp \\ &= \frac{n + 1}{2n + 1}. \end{aligned}$$

Consequently,

$$\mathbb{P}(T_{\text{next}}^n \mid T^n) \longrightarrow \frac{1}{2}.$$

At first this may seem surprising. We have seen more and more tails, and the posterior is

concentrating closer and closer to $p = 0$, yet the probability of seeing another equally long run of tails does not converge to one.

There is no contradiction. The posterior scale of p is $1/n$, while the number of future tosses under consideration is also n . Indeed, under the posterior,

$$np \xrightarrow{d} Z, \quad Z \sim \text{Exp}(1).$$

For p of order $1/n$,

$$(1 - p)^n \approx e^{-np},$$

and therefore

$$\mathbb{E}[(1 - p)^n \mid T^n] \longrightarrow \mathbb{E}[e^{-Z}] = \frac{1}{2}.$$

Thus the posterior is learning that p is small, but the prediction problem is simultaneously becoming harder: we are asking for no heads over a future horizon that grows at exactly the reciprocal scale of p .

What if the coin can be exactly incapable of producing heads?

The uniform prior has another important feature:

$$\mathbb{P}(p = 0) = 0.$$

No matter how many tails we observe, the posterior probability of the hypothesis $p = 0$ therefore remains zero. If $p = 0$ is a scientifically meaningful possibility, we should give it positive prior probability.

Consider instead the spike-and-uniform prior

$$\pi(dp) = w \delta_0(dp) + (1 - w) \mathbf{1}_{(0,1)}(p) dp, \quad 0 < w < 1.$$

Thus, with prior probability w , the coin is exactly incapable of producing a head; otherwise p is uniformly distributed on $(0, 1)$.

The probability of observing n tails under this prior is

$$\mathbb{P}(T^n) = w + (1 - w) \int_0^1 (1 - p)^n dp = w + \frac{1 - w}{n + 1}.$$

Hence the posterior probability of the point hypothesis $p = 0$ is

$$\mathbb{P}(p = 0 \mid T^n) = \frac{w}{w + \frac{1-w}{n+1}} = \frac{w(n+1)}{1 + nw}.$$

For every fixed $w > 0$,

$$\mathbb{P}(p = 0 \mid T^n) \longrightarrow 1.$$

The predictive probability of another n tails can be obtained particularly simply from

$$\mathbb{P}(T_{\text{next}}^n \mid T^n) = \frac{\mathbb{P}(T^{2n})}{\mathbb{P}(T^n)}.$$

Since

$$\mathbb{P}(T^{2n}) = w + \frac{1-w}{2n+1},$$

we obtain

$$\mathbb{P}(T_{\text{next}}^n \mid T^n) = \frac{w + \frac{1-w}{2n+1}}{w + \frac{1-w}{n+1}}.$$

Equivalently,

$$\mathbb{P}(T_{\text{next}}^n \mid T^n) = \frac{(n+1)(1+2nw)}{(2n+1)(1+nw)}.$$

For every fixed $w > 0$,

$$\mathbb{P}(T_{\text{next}}^n \mid T^n) \longrightarrow 1.$$

Thus there is a striking qualitative difference:

$$\lim_{n \rightarrow \infty} \mathbb{P}(T_{\text{next}}^n \mid T^n) = \begin{cases} \frac{1}{2}, & w = 0, \\ 1, & w > 0. \end{cases}$$

The reason is Bayesian model selection. Under the hypothesis $p = 0$, the probability of n tails is 1, whereas under the uniform alternative it is

$$\frac{1}{n+1}.$$

Thus the Bayes factor in favor of $p = 0$ against the uniform alternative is

$$\text{BF}_{0,1} = n+1.$$

No matter how small the fixed prior probability $w > 0$ is, sufficiently many consecutive tails eventually overwhelm the prior odds and drive the posterior probability of $p = 0$ toward one.

Takeaway. A continuous prior can learn that p is arbitrarily close to zero, but it can never learn that p is *exactly* zero if it assigned zero prior probability to that possibility. Giving even a tiny positive prior mass to $p = 0$ changes the asymptotic prediction dramatically: the probability of another run of n tails changes from a limiting value of $1/2$ to a limiting value of 1.

This illustrates a general Bayesian principle: if an exact hypothesis is scientifically meaningful and we want the data to be able to select it, then the prior must assign that hypothesis positive probability.

can talk about mixture of betas as in the paper [diaconis ylvisaker \(86\)](#)

4 Survey Sampling and the Likelihood Principle

Another illuminating example of the likelihood principle comes from finite-population survey sampling. The example is closely related to the stopping-rule phenomenon discussed earlier: two sampling schemes can produce exactly the same observed data and proportional likelihoods, yet lead to different frequentist answers.

The setup

Suppose that a finite population consists of $N = 10$ units with unknown values

$$Y_1, \dots, Y_{10}.$$

Our goal is to estimate the finite-population mean

$$\mu_{10} = \frac{1}{10} \sum_{j=1}^{10} Y_j.$$

We take $n = 3$ draws with replacement. Let

$$I_1, I_2, I_3 \in \{1, \dots, 10\}$$

denote the labels of the sampled units.

We compare two sampling schemes.

1. **Uniform sampling.** On each draw,

$$P(I_k = j) = \frac{1}{10}, \quad j = 1, \dots, 10.$$

2. **Unequal-probability sampling.** On each draw,

$$P(I_k = j) = \pi_j, \quad \pi_j > 0, \quad \sum_{j=1}^{10} \pi_j = 1.$$

The probabilities $\pi_1, \dots, \pi_{10}$ are fixed in advance and do not depend on the unknown population values $Y_1, \dots, Y_{10}$.

Suppose that in either experiment the realized sample is

$$(I_1, I_2, I_3) = (2, 5, 8),$$

and we observe

$$(Y_2, Y_5, Y_8) = (y_2, y_5, y_8).$$

Thus the complete observed data are

$$\mathcal{D} = ((2, y_2), (5, y_5), (8, y_8)).$$

We now ask: should our inference about the unknown population, and hence about μ_{10} , depend on which of the two sampling schemes was used?

The design-based frequentist answer

In classical survey sampling, the population values

$$Y_1, \dots, Y_{10}$$

are regarded as fixed. The randomness comes entirely from the sampling design.

Under uniform sampling, the ordinary sample mean

$$\hat{\mu}_{\text{unif}} = \frac{1}{3} \sum_{k=1}^3 Y_{I_k}$$

is unbiased for the finite-population mean, since

$$\mathbb{E}_{\text{design}}[Y_{I_k}] = \frac{1}{10} \sum_{j=1}^{10} Y_j = \mu_{10}.$$

For the particular observed sample,

$$\hat{\mu}_{\text{unif}} = \frac{y_2 + y_5 + y_8}{3}.$$

Under unequal-probability sampling, the ordinary sample mean is generally not unbiased for μ_{10} . The Horvitz–Thompson estimator is

$$\hat{\mu}_{\text{HT}} = \frac{1}{3} \sum_{k=1}^3 \frac{Y_{I_k}}{10\pi_{I_k}}.$$

Indeed,

$$\mathbb{E}_{\text{design}} \left[\frac{Y_{I_k}}{10\pi_{I_k}} \right] = \sum_{j=1}^{10} \pi_j \frac{Y_j}{10\pi_j} = \frac{1}{10} \sum_{j=1}^{10} Y_j = \mu_{10}.$$

For the same observed sample,

$$\hat{\mu}_{\text{HT}} = \frac{1}{3} \left(\frac{y_2}{10\pi_2} + \frac{y_5}{10\pi_5} + \frac{y_8}{10\pi_8} \right).$$

Thus the two experiments can produce exactly the same observed data,

$$\mathcal{D} = ((2, y_2), (5, y_5), (8, y_8)),$$

yet the design-based frequentist gives different estimates:

$$\frac{y_2 + y_5 + y_8}{3} \quad \text{versus} \quad \frac{1}{3} \left(\frac{y_2}{10\pi_2} + \frac{y_5}{10\pi_5} + \frac{y_8}{10\pi_8} \right).$$

The reason is that unbiasedness is defined by averaging over all samples that *could have been observed* under the sampling design. The sampling probabilities therefore play an essential role in the frequentist analysis.

The Bayesian answer

Let us now formulate the same problem from a Bayesian point of view.

The unknown parameter. The unknown parameter is the entire finite-population vector

$$Y = (Y_1, \dots, Y_{10}) \in \mathbb{R}^{10}.$$

There is no superpopulation parameter in this formulation. The quantity of interest

$$\mu_{10}(Y) = \frac{1}{10} \sum_{j=1}^{10} Y_j$$

is simply a function of the unknown parameter Y .

The prior. A Bayesian places a probability distribution directly on the unknown finite population:

$$Y = (Y_1, \dots, Y_{10}) \sim \Pi.$$

Here Π can be any joint probability distribution on $\mathbb{R}^{10}$. For example, a simple prior would be

$$Y_1, \dots, Y_{10} \stackrel{\text{iid}}{\sim} N(0, \tau^2),$$

although neither independence nor normality is needed for the argument below.

The observation. The observed data consist of both the labels of the sampled units and their observed values. In our example,

$$\mathcal{D} = ((2, y_2), (5, y_5), (8, y_8)).$$

Thus we have learned that

$$Y_2 = y_2, \quad Y_5 = y_5, \quad Y_8 = y_8,$$

while

$$Y_1, Y_3, Y_4, Y_6, Y_7, Y_9, Y_{10}$$

remain unknown.

The likelihood. To avoid technicalities associated with conditioning on an exact value of a continuous random variable, suppose for the moment that the Y_j 's take values in a discrete space. The same conclusion holds in the continuous case using densities and regular conditional distributions.

Consider a possible finite population

$$Y = (z_1, \dots, z_{10}).$$

The likelihood is the probability, as a function of z , of obtaining the data that were actually observed.

Under uniform sampling,

$$\begin{aligned} L_{\text{unif}}(z) &= P(\mathcal{D} \mid Y = z) \\ &= P(I_1 = 2, I_2 = 5, I_3 = 8) \mathbf{1}\{z_2 = y_2, z_5 = y_5, z_8 = y_8\} \\ &= \left(\frac{1}{10}\right)^3 \mathbf{1}\{z_2 = y_2, z_5 = y_5, z_8 = y_8\}. \end{aligned}$$

Under unequal-probability sampling,

$$\begin{aligned} L_{\pi}(z) &= P(\mathcal{D} \mid Y = z) \\ &= \pi_2 \pi_5 \pi_8 \mathbf{1}\{z_2 = y_2, z_5 = y_5, z_8 = y_8\}. \end{aligned}$$

Therefore

$$\boxed{L_{\text{unif}}(z) \propto L_{\pi}(z).}$$

The two likelihoods differ only by a multiplicative constant that does not depend on the unknown population z .

The posterior. Bayes' rule gives

$$\Pi(dz \mid \mathcal{D}) \propto L(z) \Pi(dz).$$

Under uniform sampling,

$$\Pi_{\text{unif}}(dz \mid \mathcal{D}) \propto \left(\frac{1}{10}\right)^3 \mathbf{1}\{z_2 = y_2, z_5 = y_5, z_8 = y_8\} \Pi(dz).$$

Since $(1/10)^3$ is constant with respect to the unknown z , it disappears upon normalization.

Similarly, under unequal-probability sampling,

$$\Pi_{\pi}(dz \mid \mathcal{D}) \propto \pi_2\pi_5\pi_8 \mathbf{1}\{z_2 = y_2, z_5 = y_5, z_8 = y_8\} \Pi(dz),$$

and the factor $\pi_2\pi_5\pi_8$ also disappears upon normalization.

Consequently,

$$\boxed{\Pi_{\text{unif}}(dz \mid \mathcal{D}) = \Pi_{\pi}(dz \mid \mathcal{D}).}$$

Indeed, under either sampling scheme, the posterior is simply the prior conditioned on what we have learned about the finite population:

$$\boxed{\Pi(dz \mid \mathcal{D}) = \Pi(dz \mid Y_2 = y_2, Y_5 = y_5, Y_8 = y_8).}$$

Consequently, the posterior distribution of

$$\mu_{10} = \frac{1}{10} (Y_1 + y_2 + Y_3 + Y_4 + y_5 + Y_6 + Y_7 + y_8 + Y_9 + Y_{10})$$

is exactly the same under the two sampling schemes.

Connection with the likelihood principle

This example makes the likelihood principle particularly transparent. The two experiments have different sampling distributions. For example, the probability of selecting the sequence $(2, 5, 8)$ is

$$\left(\frac{1}{10}\right)^3$$

under uniform sampling and

$$\pi_2\pi_5\pi_8$$

under unequal-probability sampling.

But once the sample $(2, 5, 8)$ has actually been observed, these probabilities are merely constants with respect to the unknown finite population Y . Thus the two experiments give proportional likelihoods:

$$L_{\text{unif}}(Y) \propto L_{\pi}(Y).$$

According to the likelihood principle, two experiments producing proportional likelihoods should lead to the same inference about the unknown parameter. Bayesian inference satisfies this automatically: the sampling probabilities cancel from Bayes' rule, and the posterior distribution of the finite population, and hence of its mean, is identical under the two designs.

The classical design-based frequentist analysis does not satisfy this principle. Its criterion of unbiasedness is defined by averaging over repeated samples generated by the sampling design. As a result, probabilities attached to samples that *could have been observed* affect the inference

made from the sample that was actually observed. The same realized data can therefore lead to the ordinary sample mean under one design and an inverse-probability-weighted estimator under another.

This is closely analogous to the stopping-rule phenomenon. In both examples, the Bayesian conditions on what was actually observed and uses the resulting likelihood, whereas the frequentist sampling distribution also depends on outcomes that could have occurred but did not.

Remark 4.1 (When the sampling design cannot be ignored). *The argument above relies crucially on the sampling probabilities $\pi_1, \dots, \pi_{10}$ being fixed independently of the unknown population values. If the probability that a unit is sampled depends on its value, then*

$$P(I_k = j \mid Y)$$

is itself a function of the unknown population Y . The sampling mechanism then contributes non-constant information to the likelihood and cannot be discarded from the Bayesian analysis.

Thus the conclusion is not that Bayesian inference always ignores the sampling mechanism, but rather that a noninformative sampling mechanism disappears from the posterior once we condition on the observed sample.

4.1 Basu's Elephant Example

A famous example due to Debabrata Basu shows, in a deliberately extreme way, how strange design-based unbiased estimation can become.

A circus has 50 elephants with unknown weights

$$Y_1, \dots, Y_{50},$$

and the owner wants to estimate their total weight

$$T = \sum_{i=1}^{50} Y_i.$$

Weighing all 50 elephants is inconvenient, so he would like to weigh only one.

From records taken three years earlier, the owner knows that one elephant, *Sambo*, had approximately the average weight of the herd. The trainer believes that Sambo is still reasonably representative. The owner therefore proposes the natural procedure: weigh Sambo and estimate

$$T \quad \text{by} \quad 50Y_S,$$

where Y_S denotes Sambo's current weight.

A design-based statistician objects. Sambo has been selected purposively rather than through a randomized sampling design. The owner and the statistician therefore agree on the following

compromise:

$$P(I = S) = \frac{99}{100},$$

while each of the remaining 49 elephants is selected with probability

$$P(I = i) = \frac{1}{4900}, \quad i \neq S.$$

If elephant I is selected, the Horvitz–Thompson estimator of the total is

$$\hat{T}_{\text{HT}} = \frac{Y_I}{\pi_I},$$

where $\pi_i = P(I = i)$. It is exactly design-unbiased:

$$\mathbb{E}_{\text{design}} \left[\hat{T}_{\text{HT}} \right] = \sum_{i=1}^{50} \pi_i \frac{Y_i}{\pi_i} = \sum_{i=1}^{50} Y_i = T.$$

Now suppose, as happens with probability 0.99, that Sambo is selected. The estimate is

$$\hat{T}_{\text{HT}} = \frac{Y_S}{0.99} \approx Y_S.$$

Thus, after weighing an elephant believed to have approximately the *average* weight of the herd, the Horvitz–Thompson estimator says that the *total weight of all 50 elephants* is approximately the weight of that one elephant.

How can such an estimator possibly be unbiased?

The answer is revealed by considering what happens on the rare event that one of the other elephants is selected. Suppose, for example, that *Jumbo*, a particularly large elephant, is selected. Since

$$\pi_J = \frac{1}{4900},$$

the estimate becomes

$$\hat{T}_{\text{HT}} = 4900Y_J.$$

Thus the estimated total weight of 50 elephants is 4900 times Jumbo’s weight.

There is no mathematical error. The overwhelmingly likely severe underestimate obtained when Sambo is selected is compensated, in expectation, by enormous estimates on the very rare occasions when one of the other elephants is selected. This is precisely how

$$\mathbb{E}_{\text{design}} \left[\hat{T}_{\text{HT}} \right] = T$$

is achieved.

Basu’s point was not that the Horvitz–Thompson estimator is always a bad estimator. Rather, the example shows that *design unbiasedness by itself is not a sufficient principle for sensible sta-*

tistical inference. A procedure can be exactly unbiased under repeated applications of a sampling design and nevertheless give an absurd answer for the data actually observed.

The example also makes the connection with the likelihood principle especially vivid. Suppose Sambo is selected and we observe

$$Y_S = y_S.$$

The probability 0.99 of selecting Sambo was fixed in advance and does not depend on any of the unknown elephant weights. Therefore, as a function of the unknown population Y , the likelihood has the form

$$L(Y) \propto \mathbf{1}\{Y_S = y_S\};$$

the factor 0.99 is merely a multiplicative constant.

Consequently, for a Bayesian,

$$\boxed{\Pi(dY \mid I = S, Y_S = y_S) = \Pi(dY \mid Y_S = y_S).}$$

There is no reason to divide Sambo's observed weight by 0.99. If we have prior information that Sambo is likely to be representative of the herd, that information can instead be incorporated into the prior distribution on

$$(Y_1, \dots, Y_{50}).$$

Sambo's observed weight can then provide information about the unobserved elephants and hence about the total T .

Basu's deliberately absurd example therefore highlights the same distinction as the preceding survey-sampling discussion. The design-based frequentist evaluates an estimator by averaging over samples that *could have been observed*. In this example, the rare possibility of selecting Jumbo is essential to the unbiasedness of the estimate obtained when Sambo is actually selected.

The Bayesian, by contrast, conditions on the sample that *was actually observed*. When the sampling design is independent of the unknown population values, the sampling probabilities carry no additional information about those values and disappear from the posterior.

Remark 4.2 (A prior expressing that Sambo is representative). *The Bayesian formulation also gives a precise way to incorporate the owner's prior knowledge that Sambo is a representative elephant. Ideally, we would like a prior under which Sambo's weight is close to the actual finite-population average*

$$\bar{Y} = \frac{1}{50} \sum_{i=1}^{50} Y_i.$$

Here is one simple construction. First draw the overall average weight

$$M \sim N(m_0, v_0).$$

Next draw

$$\delta \sim N(0, \tau^2),$$

where τ^2 is small, and set

$$Y_S = M + \delta.$$

Thus δ measures how far Sambo is from the average.

For the remaining 49 elephants, draw

$$U_i \stackrel{\text{iid}}{\sim} N(0, \sigma^2), \quad i \neq S,$$

and let

$$\bar{U} = \frac{1}{49} \sum_{i \neq S} U_i.$$

Define

$$Y_i = M + (U_i - \bar{U}) - \frac{\delta}{49}, \quad i \neq S.$$

By construction,

$$\sum_{i=1}^{50} (Y_i - M) = \delta + \sum_{i \neq S} (U_i - \bar{U}) - \delta = 0.$$

Hence

$$\boxed{\bar{Y} = \frac{1}{50} \sum_{i=1}^{50} Y_i = M.}$$

Moreover,

$$\boxed{Y_S - \bar{Y} = \delta \sim N(0, \tau^2).}$$

Thus τ^2 has a direct interpretation: it measures our uncertainty about how representative Sambo is. Taking τ^2 small expresses the belief that Sambo's weight is very close to the actual average weight of the 50 elephants. The parameter σ^2 , on the other hand, controls the ordinary variation in weight among the remaining elephants.